Formal Proof of Asymptotic Predictive Risk Upper Bound in High-Dimensional Transfer Learning with Missing Features

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1 Introduction and Problem Setup

We analyze a transfer learning framework consisting of a target site t and a source site s. For each site $m \in \{t, s\}$, the true data-generating process satisfies the linear model:

$$y^{(m)} = x^\top \beta^{(m)} + \varepsilon, \tag{1}$$

where $\beta^{(m)} \in \mathbb{R}^p$ is a random coefficient vector, $x \sim N(0, \Sigma)$, and $\varepsilon \sim N(0, \sigma^2)$ is independent of x.

At the target site, we only observe a subvector $x_{\mathcal{R}(t)}$ indexed by $\mathcal{R}(t) \subseteq [p]$. Let $p_t = |\mathcal{R}(t)|$ and let $[\mathcal{R}(t)]^c = [p] \setminus \mathcal{R}(t)$ denote the unobserved feature indices. Let $X^{(t)}_{\mathcal{R}(t)} \in \mathbb{R}^{n_t \times p_t}$ and $Y^{(t)} \in \mathbb{R}^{n_t}$ denote the observed design matrix and response vector at the target site, respectively.

Given an estimator $\hat{\beta}^{(s)}$ of the source-site coefficient vector, the transfer learning estimator $\hat{\omega}^{(t)}_{\mathrm{TL}}$ for the target site is defined as:

$$\hat{\omega}^{(t)}_{\mathrm{TL}} := \arg\min_{\omega \in \mathbb{R}^{p_t}} \left[\frac{1}{n_t} \|Y^{(t)} - X^{(t)}_{\mathcal{R}(t)} \omega\|_2^2 + \lambda_\omega \|\omega - \hat{\omega}^{(t)}_s\|_2^2 \right], \tag{2}$$

where $\hat{\omega}^{(t)}_s = \Phi^{(t)} \hat{\beta}^{(s)}$. The projection matrix $\Phi^{(t)} \in \mathbb{R}^{p_t \times p}$ is defined such that its blocks satisfy:

$$[\Phi^{(t)}]_{i,j \in \mathcal{R}(t)} = I_{p_t}, \quad [\Phi^{(t)}]_{i \in \mathcal{R}(t), j \in [\mathcal{R}(t)]^c} = \left(\Sigma^{(t)}_{\mathcal{R}(t)}\right)^{-1} \Sigma^{(t)}_{\mathcal{R}(t),[\mathcal{R}(t)]^c}. \tag{3}$$

For a new independent test observation $(\tilde{x}_{\mathcal{R}(t)}, \tilde{y}^{(t)})$ from the target site, the conditional predictive risk is:

$$r^{\mathrm{pred}}\big(\hat{\omega}^{(t)}_{\mathrm{TL}}\big) = \mathbb{E}\left[\big(\tilde{y}^{(t)} - \tilde{x}^\top_{\mathcal{R}(t)} \hat{\omega}^{(t)}_{\mathrm{TL}}\big)^2 \;\middle|\; X^{(t)}_{\mathcal{R}(t)} \right]. \tag{4}$$

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2 Regularity Conditions and Assumptions

To establish the exact high-dimensional asymptotic limits, we invoke the following formal assumptions matching the conventions of Dobriban and Wager (2018) and Gu, Han, and Duan (2025):

Assumption 1 (High-Dimensional Regime)**.** *As $n_t \to \infty$, both $p \to \infty$ and $p_t \to \infty$ such that:*

$$\frac{p}{n_t} \to \gamma \in (0, \infty) \quad \textit{and} \quad \frac{p_t}{n_t} \to \eta \in (0, \infty). \tag{5}$$

Assumption 2 (Covariance Spectral Bounds)**.** *The population covariance matrix Σ is symmetric positive definite, and there exist constants c_0, C_0 such that uniformly for all p:*

$$0 < c_0 \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq C_0 < \infty. \tag{6}$$

Assumption 3 (Random Regression Coefficients)**.** *For $m \in \{s, t\}$, the random vectors $\beta^{(m)}$ satisfy $\mathbb{E}[\beta^{(m)}] = 0$, $\mathrm{Cov}(\beta^{(m)}) = \frac{\alpha_m^2}{p} I_p$ ($\alpha_m \neq 0$), and $\mathrm{Cov}(\beta^{(t)}, \beta^{(s)}) = \frac{\rho \alpha_t \alpha_s}{p} I_p$ for $\rho \in (0, 1]$.*

Assumption 4 (Source Estimation Bounds)**.** *The source estimator $\hat{\beta}^{(s)}$ satisfies:*

$$\lambda_{\max}\left(\mathbb{E}\left[(\hat{\beta}^{(s)} - \beta^{(s)})(\hat{\beta}^{(s)} - \beta^{(s)})^\top \right] \right) \leq \frac{C^U}{p}, \tag{7}$$

uniformly as $(n_t, p) \to \infty$ for a positive constant C^U.

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3 Formal Mathematical Proof

3.1 Step 1: Structural Decomposition of Missing Features

We partition the target covariate vector x and the full target coefficient vector $\beta^{(t)}$ into observed and unobserved coordinates:

$$x = \begin{pmatrix} x_{\mathcal{R}(t)} \\ x_{[\mathcal{R}(t)]^c} \end{pmatrix}, \quad \beta^{(t)} = \begin{pmatrix} \beta^{(t)}_{\mathcal{R}(t)} \\ \beta^{(t)}_{[\mathcal{R}(t)]^c} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{\mathcal{R}(t)} & \Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c} \\ \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)} & \Sigma_{[\mathcal{R}(t)]^c} \end{pmatrix}. \tag{8}$$

Using the properties of the joint multivariate normal distribution, the conditional expectation of the unobserved features given the observed features is:

$$\mathbb{E}\left[x_{[\mathcal{R}(t)]^c} \mid x_{\mathcal{R}(t)} \right] = \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)} \left(\Sigma_{\mathcal{R}(t)} \right)^{-1} x_{\mathcal{R}(t)}. \tag{9}$$

We rewrite the full model equation using this orthogonal projection:

$$\begin{aligned} y^{(t)} &= x_{\mathcal{R}(t)}^{\top}\beta_{\mathcal{R}(t)}^{(t)} + x_{[\mathcal{R}(t)]^c}^{\top}\beta_{[\mathcal{R}(t)]^c}^{(t)} + \varepsilon \\ &= x_{\mathcal{R}(t)}^{\top}\left(\beta_{\mathcal{R}(t)}^{(t)} + \left(\Sigma_{\mathcal{R}(t)}\right)^{-1}\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c}\beta_{[\mathcal{R}(t)]^c}^{(t)}\right) + \xi^{(t)} + \varepsilon, \end{aligned} \tag{10}$$

where $\xi^{(t)} = \left(x_{[\mathcal{R}(t)]^c} - \mathbb{E}[x_{[\mathcal{R}(t)]^c} \mid x_{\mathcal{R}(t)}]\right)^{\top}\beta_{[\mathcal{R}(t)]^c}^{(t)}$. By construction, $\xi^{(t)}$ is independent of $x_{\mathcal{R}(t)}$, with conditional variance:

$$\mathbb{E}\left[(\xi^{(t)})^2 \;\middle|\; \beta^{(t)}\right] = \left(\beta_{[\mathcal{R}(t)]^c}^{(t)}\right)^{\top}\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}\beta_{[\mathcal{R}(t)]^c}^{(t)}, \tag{11}$$

where $\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)} = \Sigma_{[\mathcal{R}(t)]^c} - \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)}\left(\Sigma_{\mathcal{R}(t)}\right)^{-1}\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c}$. Notice that by definition of $\Phi^{(t)}$, the implied low-dimensional true target coefficient vector $\omega_0^{(t)}$ matches the term in parentheses:

$$\omega_0^{(t)} := \Phi^{(t)}\beta^{(t)} = \beta_{\mathcal{R}(t)}^{(t)} + \left(\Sigma_{\mathcal{R}(t)}\right)^{-1}\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c}\beta_{[\mathcal{R}(t)]^c}^{(t)}. \tag{12}$$

Thus, the sample training response vector can be exactly represented as:

$$Y^{(t)} = X_{\mathcal{R}(t)}^{(t)}\omega_0^{(t)} + \mathcal{E}^{(t)}, \quad \mathcal{E}^{(t)} = X_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}^{(t)}\beta_{[\mathcal{R}(t)]^c}^{(t)} + \varepsilon^{(t)}, \tag{13}$$

where the effective noise vector $\mathcal{E}^{(t)}$ is independent of $X_{\mathcal{R}(t)}^{(t)}$ and satisfies $\mathbb{E}[\mathcal{E}^{(t)} \mid X_{\mathcal{R}(t)}^{(t)}] = 0$.

3.2 Step 2: Estimation Error Representation via Resolvents

Let $\hat{\Sigma}_t = \frac{1}{n_t}(X_{\mathcal{R}(t)}^{(t)})^{\top}X_{\mathcal{R}(t)}^{(t)}$ denote the empirical covariance matrix, and define the resolvent operator $R(\lambda_\omega) = (\hat{\Sigma}_t + \lambda_\omega I_{p_t})^{-1}$. Finding the first-order optimality condition of the transfer learning objective function gives:

$$\hat{\omega}_{\mathrm{TL}}^{(t)} = R(\lambda_\omega)\left(\frac{1}{n_t}(X_{\mathcal{R}(t)}^{(t)})^{\top}Y^{(t)} + \lambda_\omega\hat{\omega}_s^{(t)}\right). \tag{14}$$

Substituting $Y^{(t)} = X_{\mathcal{R}(t)}^{(t)}\omega_0^{(t)} + \mathcal{E}^{(t)}$ yields:

$$\hat{\omega}_{\mathrm{TL}}^{(t)} = R(\lambda_\omega)\hat{\Sigma}_t\omega_0^{(t)} + R(\lambda_\omega)\frac{1}{n_t}(X_{\mathcal{R}(t)}^{(t)})^{\top}\mathcal{E}^{(t)} + \lambda_\omega R(\lambda_\omega)\hat{\omega}_s^{(t)}. \tag{15}$$

Applying the algebraic identity $R(\lambda_\omega)\hat{\Sigma}_t = I_{p_t} - \lambda_\omega R(\lambda_\omega)$, we isolate the estimation error vector:

$$\hat{\omega}_{\mathrm{TL}}^{(t)} - \omega_0^{(t)} = \lambda_\omega R(\lambda_\omega)(\hat{\omega}_s^{(t)} - \omega_0^{(t)}) + R(\lambda_\omega)\frac{1}{n_t}(X_{\mathcal{R}(t)}^{(t)})^{\top}\mathcal{E}^{(t)}. \tag{16}$$

3.3 Step 3: Predictive Risk Decomposition

Conditioned on the training observations, the predictive risk of $\hat{\omega}_{\mathrm{TL}}^{(t)}$ on an independent test pair $(\tilde{x}_{\mathcal{R}(t)}, \tilde{y}^{(t)})$ is decomposed via the standard bias-variance framework. Letting $\Sigma_t = \Sigma_{\mathcal{R}(t)}$, we write:

$$r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) = \sigma_{\mathcal{E}}^2 + \mathbb{E}\left[\left\|\hat{\omega}_{\mathrm{TL}}^{(t)} - \omega_0^{(t)}\right\|_{\Sigma_t}^2 \;\middle|\; X_{\mathcal{R}(t)}^{(t)}\right], \tag{17}$$

where $\sigma_{\mathcal{E}}^2 = \sigma^2 + \mathbb{E}[(\xi^{(t)})^2 \mid \beta^{(t)}]$. Substituting the explicit error representation into the semi-norm $\|\cdot\|_{\Sigma_t}^2$, the conditional cross-term vanishes because $\mathbb{E}[\mathcal{E}^{(t)} \mid X_{\mathcal{R}(t)}^{(t)}] = 0$. Hence, the risk splits cleanly into a Bias component (B) and a Variance component (V):

$$\mathbb{E}[R_t] = B + V, \tag{18}$$

$$B = \lambda_\omega^2 \mathbb{E}\left[(\hat{\omega}_s^{(t)} - \omega_0^{(t)})^\top R(\lambda_\omega)\Sigma_t R(\lambda_\omega)(\hat{\omega}_s^{(t)} - \omega_0^{(t)})\right], \tag{19}$$

$$V = \frac{1}{n_t^2}\mathbb{E}\left[(\mathcal{E}^{(t)})^\top X_{\mathcal{R}(t)}^{(t)} R(\lambda_\omega)\Sigma_t R(\lambda_\omega)(X_{\mathcal{R}(t)}^{(t)})^\top \mathcal{E}^{(t)}\right]. \tag{20}$$

3.4 Step 4: Deterministic Equivalents and Spectral Limits

Following Dobriban and Wager (2018), the random resolvent matrix $R(\lambda_\omega)$ can be approximated by its deterministic equivalent $B(\lambda_\omega) = (v(\lambda_\omega)\Sigma_t + \lambda_\omega I_{p_t})^{-1}$, where $v(\lambda_\omega) > 0$ is the unique solution to the fixed-point equation:

$$v(\lambda_\omega) = \frac{1}{1 + \frac{1}{n_t}\operatorname{Tr}\left(\Sigma_t(v(\lambda_\omega)\Sigma_t + \lambda_\omega I_{p_t})^{-1}\right)} = \frac{1}{1 + \eta\Psi(v(\lambda_\omega), \lambda_\omega)}, \tag{21}$$

with $\Psi(v, \lambda) = \frac{1}{p_t}\operatorname{Tr}(\Sigma_t(v\Sigma_t + \lambda I_{p_t})^{-1})$.

3.4.1 Evaluating the Variance Component

Evaluating the expectation over the effective noise vector yields $\mathbb{E}[\mathcal{E}^{(t)}(\mathcal{E}^{(t)})^\top \mid X_{\mathcal{R}(t)}^{(t)}] = \sigma_{\mathcal{E}}^2 I_{n_t}$. Substituting this into V yields:

$$V = \frac{\sigma_{\mathcal{E}}^2}{n_t}\operatorname{Tr}\left(\hat{\Sigma}_t R(\lambda_\omega)\Sigma_t R(\lambda_\omega)\right). \tag{22}$$

Applying the random matrix trace limits from Dobriban and Wager (2018), we obtain:

$$V \xrightarrow{p} \sigma_{\mathcal{E}}^2 \cdot \frac{\eta v(\lambda_\omega)^2\theta(\lambda_\omega)}{1 - \eta v(\lambda_\omega)^2\theta(\lambda_\omega)}, \tag{23}$$

where $\theta(\lambda_\omega) = \frac{1}{p_t}\operatorname{Tr}\left(\Sigma_t^2(v(\lambda_\omega)\Sigma_t + \lambda_\omega I_{p_t})^{-2}\right)$.

3.4.2 Evaluating and Bounding the Bias Component

Let $\Delta_\omega = \hat{\omega}_s^{(t)} - \omega_0^{(t)}$. Using the definition of the estimators, we expand the anchor error vector:

$$\Delta_\omega = \Phi^{(t)}(\hat{\beta}^{(s)} - \beta^{(t)}) = \Phi^{(t)}(\hat{\beta}^{(s)} - \beta^{(s)}) + \Phi^{(t)}(\beta^{(s)} - \beta^{(t)}). \tag{24}$$

Taking the outer product expectation under Assumptions 3 and 4:

$$\mathbb{E}[(\beta^{(s)} - \beta^{(t)})(\beta^{(s)} - \beta^{(t)})^\top] = \frac{\alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t}{p} I_p, \tag{25}$$

$$\lambda_{\max}\left(\mathbb{E}[(\hat{\beta}^{(s)} - \beta^{(s)})(\hat{\beta}^{(s)} - \beta^{(s)})^\top]\right) \leq \frac{C^U}{p}. \tag{26}$$

Combining these bounds yields the matrix inequality:

$$\mathbb{E}[\Delta_\omega \Delta_\omega^\top] \preceq \frac{\alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t + C^U}{p} \Phi^{(t)}(\Phi^{(t)})^\top. \tag{27}$$

We bound the spectrum of the outer product $\Phi^{(t)}(\Phi^{(t)})^\top$. By its block form construction:

$$\Phi^{(t)}(\Phi^{(t)})^\top = I_{p_t} + \left(\Sigma_{\mathcal{R}(t)}\right)^{-1} \Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c} \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)} \left(\Sigma_{\mathcal{R}(t)}\right)^{-1}. \tag{28}$$

Using the Schur complement relationship $\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c} \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)} \preceq \Sigma_{\mathcal{R}(t)}$, it follows from Assumption 2 that:

$$\Phi^{(t)}(\Phi^{(t)})^\top \preceq \left(1 + \lambda_{\max}(\Sigma)\lambda_{\min}^{-1}(\Sigma)\right) I_{p_t}. \tag{29}$$

Substituting the deterministic equivalent $B(\lambda_\omega)$ for $R(\lambda_\omega)$, the bias term admits the upper bound:

$$B \leq \lambda_\omega^2 \frac{\alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t + C^U}{p} \left(1 + \frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}\right) \operatorname{Tr}\left((v(\lambda_\omega)\Sigma_t + \lambda_\omega I_{p_t})^{-2}\Sigma_t\right). \tag{30}$$

Using Assumption 1 to replace the dimension ratio $\frac{p_t}{p} \to \frac{\eta}{\gamma}$, we normalize the trace to dimension p_t:

$$B \leq \lambda_\omega^2 \frac{\eta}{\gamma} M \cdot \frac{1}{p_t} \operatorname{Tr}\left(\Sigma_t(v(\lambda_\omega)\Sigma_t + \lambda_\omega I_{p_t})^{-2}\right) + o_p(1), \tag{31}$$

where $M = (\alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t + C^U)\left(1 + \lambda_{\max}(\Sigma)\lambda_{\min}^{-1}(\Sigma)\right)$.

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4 Asymptotic Upper Bound and Optimal λ_ω

Combining the components yields the final asymptotic upper bound for the predictive risk.

Theorem 1. *Under Assumptions 1–4, the predictive risk of the transfer learning estimator satisfies:*

$$r^{\text{pred}}\big(\hat{\omega}_{\text{TL}}^{(t)}\big) \leq \sigma_{\mathcal{E}}^2 + \left[\sigma_{\mathcal{E}}^2 \frac{\eta v(\lambda_\omega)^2 \theta(\lambda_\omega)}{1 - \eta v(\lambda_\omega)^2 \theta(\lambda_\omega)} + \lambda_\omega^2 \frac{\eta}{\gamma} M \cdot \Psi_2(\lambda_\omega)\right] + o_p(1), \quad (32)$$

where $\Psi_2(\lambda_\omega) = \frac{1}{p_t} \operatorname{Tr}\left(\Sigma_t (v(\lambda_\omega)\Sigma_t + \lambda_\omega I_{p_t})^{-2}\right)$ *and* $M = (\alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t + C^U)\left(1 + \frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}\right)$.

The optimal tuning parameter λ_ω^* minimizing this high-dimensional risk bound balances the effective noise against the structural anchor bias, yielding:

$$\lambda_\omega^* = \frac{\gamma \sigma_{\mathcal{E}}^2}{\eta M} = \frac{\gamma\left(\sigma^2 + \alpha_t^2 \cdot \frac{p - p_t}{p} \bar{\lambda}(\Sigma_{[\mathcal{R}(t)]^c | \mathcal{R}(t)})\right)}{\eta(\alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t + C^U)\left(1 + \frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}\right)}, \quad (33)$$

where $\bar{\lambda}(\cdot)$ denotes the average eigenvalue of the conditional sub-covariance matrix.